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THE ORGANIZATION OF HEREDITARY MAZE-BRIGHTNESS AND MAZE-DULLNESS* †

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It is obvious that primary credit is due to Professor Robert C. Tryon, whose care in planning and conducting his own famous experiment is responsible for marked differences which occur in the performances of Brights and Dulls. The writer is indebted to him for suggesting the initial plan of this study, as well as for generously donating its subjects.

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I. INTRODUCTION

A. Purpose

The rats of Tryon's genetic strains have been selectively bred on the basis of their error scores in a complex alley maze, under conditions of rigorous environmental control, until two groups have resulted whose hereditary abilities to learn the maze are almost completely separate (18).

The Tryon study represents a significant experimental approach to the problem of heredity and environment. The trait of maze-learning ability has been generally accepted as the best available index of rat "intelligence," even though its psychological nature remains a matter of wide disagreement. In the results of this experiment it has been clearly shown that such a complex trait has a controllable hereditary basis. Specifically, the experiment succeeded in separating the genetic determiners into two sets, one favorable to maze-brightness and the other to maze-dullness. The gradual nature of separation of the strains through succeeding generations provides evidence that the independent factors involved are numerous (16).

These strains of rats constitute invaluable material for further investigation. Although it is pointless to argue whether heredity is more or less important than environment in the learning abilities of individual rats, it is clear that the existing difference between mean performances of these Bright and Dull groups is due wholly to genetic causes. Since maze error scores were used as the sole criterion of selection, this difference may be regarded as relating directly to the maze-learning function. Thus, any systematic differences occurring in various other behaviors of Brights and Dulls reflect the operation of the same sets of causes, and the common elements of such behaviors may be regarded as constituting basic psychological components of the maze-learning ability.

The present study undertakes to provide an analytical description of the total organization of behavior for each of the two strains, and in terms of the analyzed components, to present a preliminary explanation of the rats' maze-learning performances. Before the study was begun, evidence had already been obtained that the basic psychological difference was not to be conceived as a simple matter of ability to learn. In testing small samples of the Brights and Dulls in an exploratory series of problems, the writer had found that the Brights' superiority tended to disappear when stimulus conditions were made sufficiently different from those of the alley maze. In one case, involving a three-dimensional under-water maze, a slight but reliable difference was obtained in favor of the Dulls. Although results of these tests were

not conclusive, they were sufficient to indicate in a preliminary way that differences in a general factor of learning were either very slight or nonexistent. It might have been supposed, in line with conventional factor theory, that the results were explainable in terms of group or specific factors of learning, but this explanation did not seem tenable, either, in view of the striking consistency which both groups showed in certain of their "personality" patterns. It was observed that individual rats reacted in characteristic ways throughout widely varying problem situations. Although the responses were not systematically measured during the preliminary experiments, there were definite indications that reliable differences existed between Brights and Dulls in their emotional and motivational natures.[1] On the whole, the suggestion was strong that the groups represented distinctive personality *types,* whose contrasting degrees of maze-brightness were to be regarded as but one aspect of their different total organizations of behavior.

B. Statistical Design

In previous studies of trait organization in animals, the method generally employed has been that of intercorrelating scores by tests, where a variety of test measures has been obtained for a heterogeneous sample of the animals under study (2, 7, 13, 15, 22). Normally the data are treated by some method of factor analysis, and in recent years the concept of "trait organization" has come to be identified with this kind of experimental approach. For the present study, however, where the question of principal interest relates to differences in organization between two pure strains, the conventional approach could not be taken. The proposal that a separate analysis might be made for each strain was not seriously considered on the grounds that intercorrelations would tend generally towards zero to the extent that the strains were actually homogeneous, and also because differences which might exist between the two strains on various individual test measures would be lost in the process. More importantly, the subject of investigation is regarded as basically contradictory to the conventional assumption that trait organization is identical for all members of a given species.[2]

Based upon the above considerations, a decision was made to design the

[1] Emotionality studies of Brights and Dulls had been made previously by Tryon, Tryon, and Kuznets (20, 21).

[2] That is, that all individuals' performances in a given test involve the same set of component traits and in the same relative proportions. In multiple-factor methods this assumption is expressed by the formula, $x_a = a_1x_1 + a_2x_2 + \ldots + a_nx_n$, wherein the weights, $a_1, a_2, \ldots a_n$, may vary for other tests but in test a are constant for all individuals.

present experiment along the lines of techniques which have been employed by Stephenson (10) and by Zubin (24) for studying psychological "types" among humans. Briefly described, these techniques involve computing measures of relation among the various individuals, instead of among tests. Where a series of tests is considered to be a sample of behaviors in terms of which individuals show distinctive patterns, degrees of resemblance among individuals may be determined by intercorrelating the patterns, and as Stephenson has demonstrated (11), the intercorrelations may be treated by ordinary methods of factor analysis. The resulting factors—i.e., clusterings of variables—represent *types* of individuals.

Applied to an experiment with samples of Brights and Dulls, the method may be outlined in advance as involving the following steps: (*a*) A distribution of (comparable) scores is obtained for each rat in a series of test measures, where the series represents a sample of behaviors selected from the universe of rat behavior problems. (*b*) The correlation is computed between each pair of rats. For the total number, k, of Brights plus Dulls, the number of intercorrelations computed will be $k(k - 1)/2$. (*c*) Using some method of group factor analysis, the table of intercorrelations is analyzed for factors. Following these steps, the existence of hereditary behavior types might be considered demonstrated if factor saturations for individual members of the two strains were to provide evidence for a "Bright" factor and a "Dull" factor. A further identification of each type should then be made in terms of its characteristic pattern of behaviors.

In the course of designing the experiment, a problem was encountered initially in choosing a way to make the various test scores comparable. In raw form the scores would be obtained as numbers of errors, seconds, cage revolutions, etc., and would evidently have to be converted into units of some common scale before meaning could be attached to the distributions for individual rats. The previously suggested solutions to this problem have been either (*a*) to use only tests for which the units are already comparable in raw form, or (*b*) to convert the original scores in each test into units based upon the means and standard deviations of the experimental sample. The first solution is, of course, not applicable in the present case. The second was considered undesirable in view of the probable variation to be found among the forms of test distributions (many of these would be platykurtic or bimodal as a result of combining scores for Brights and Dulls), and also because intercorrelations among individuals are to some extent mathematically predetermined when test means and sigmas have been standardized with

reference to the experimental sample.[3] The decision was therefore made to employ a "reference" group of rats, representing the original population from which Brights and Dulls were bred, and to convert scores for rats of the experimental samples into a common scale based upon the normalized distributions of test scores for the reference group. Although complicating the experimental procedure in some respects, this method is free from the above objections and in addition, permits Brights' and Dulls' performances to be evaluated as "high" "low," or "average" on each test measure in terms which have a relatively standard meaning. It may be noted in passing that the principles of the method are identical to those which are involved in making up educational achievement profiles for school children, when the common scale is based on established "norms" for a set of tests (3).

C. Animals

The samples of Brights and Dulls were taken from the 22nd generation of the Tryon strains. They were members of a colony stock which had been removed from the regular experimental groups since the F_{19} generation, and for this reason had not been selectively bred on the basis of maze error scores since that time. They had been selected on the basis of coat color in order to prevent accidental intermixture of the two strains. The Brights had gray coats and the Dulls black.

For the reference group, a sample was chosen from a stock of rats known as "*M* by *M*" (*MxM*), so called because they were the progeny (F_7) of a median strain which had been introduced during an earlier generation by crossing members of the two pure strains. Presumably, rats bred in this way would resemble most closely the original parent population.

The three groups included a total of 40 rats. Of these, 12 were Brights, 12 were Dulls, and 10 were *MxM*'s. They were chosen from 10 litters of Brights, 10 of Dulls, and 16 of *MxM*'s (the total stock of the current generation), one rat being taken from each litter in order to provide the most representative samples obtainable with a small number of cases. Two Brights, two Dulls, and one *MxM* were initially designated to serve as spare replacements in the event that members of the experimental samples proper should be lost through sickness or accident. Only one of these five spares, G-6, was eventually included as a regular member.

The individual rats' numbers, coat color, ages, and weights are listed in Table 1. All rats were male and had pigmented eyes. As may be seen in

[3] The average intercorrelation cannot be less than the value, $-1/(k-1)$ (cf. ref. 1, p. 275), for a sample of k members, nor greater than zero.

TABLE 1

EXPERIMENTAL ANIMALS: NUMBER DESIGNATIONS, AGES, WEIGHTS, AND COAT COLOR
(All rats were males with pigmented eyes)

	No.	Coat Color	Age Begin (days)	Age End (days)	Wgt. Begin (grams)	Wt. End (grams)
Brights (F_{22})	G- 6	Gray	96	346	285	404
	G- 5	Gray	96	346	275	402
	G-10	Gray	99	349	270	420
	G-12	Gray	99	349	250	385
	G-16	Gray	98	348	290	404
	G-18	Gray	96	346	285	406
	G-24	Gray	91	341	310	444
	G-27	Gray	88	338	215	344
	G-28	Gray	76	326	275	434
	G-30	Gray	72	322	190	360
	Mean		91	341	264	401
Dulls (F_{22})	B- 3	Black	98	348	245	380
	B- 6	Black	96	346	230	316
	B- 8	Black	95	345	195	256
	B-10	Black	95	345	250	352
	B-12	Black	95	345	195	292
	B-15	Black	95	345	220	280
	B-18	Black	95	345	230	282
	B-22	Black	95	345	180	334
	B-24	Black	95	345	240	374
	B-26	Black	91	341	210	322
	Mean		95	345	220	319
M×M's (F_7)	BH- 5	B.Hood	96	346	260	380
	BH- 9	B.Hood	82	332	240	356
	BH-11	B.Hood	96	346	270	374
	BH-13	B.Hood	99	349	210	370
	BH-17	B.Hood	96	346	260	360
	BH-22	B.Hood	93	343	265	378
	BH-26	B.Hood	101	351	225	282
	BH-30	B.Hood	99	349	250	376
	BH-33	B.Hood	101	351	240	340
	BH-37	B.Hood	96	346	235	344
	BH-41	B.Hood	99	349	270	310
	BH-45	B.Hood	95	345	210	374
	BH-48	B.Hood	84	334	210	364
	BH-54	B.Hood	84	334	195	360
	BH-56	B.Hood	72	322	255	444
	Mean		93	343	240	361

the tabulated figures, the groups were essentially equal in age, but their weights show a distinct positive correlation with hereditary maze-brightness.[4] With the exception of one Bright rat who died of a lung disorder, all rats maintained excellent health throughout the experiment.

[4] An unpublished study by Tryon indicates that a weight difference in favor of the Brights is characteristic.

D. General Procedure

During a period of eight months, 30 measures of the rats' behavior were obtained in nine different apparatus situations. These are listed in Table 2 in the order, from beginning to end of the experiment, in which each apparatus was employed.

TABLE 2
List of Variables

OPEN FIELD	1. Defecations
	2. Urinations
	3. Rearings
	4. Washings
	5. Distance
TRYON MAZE	6. Errors (2-19)
	7. Time (2-19)
	8. Errors (20-21)
	9. Time (20-21)
WATER TANK	10. Errors
	11. Time
	12. Starting Time
WATER UNIT	13. Starting Time
ELEVATED PATH	14. Time
14-UNIT MAZE	15. Errors
	16. Time
	17. Starting Time
REVOLVING CAGES	18. Total Activity
	19. Pre-feeding Activity
16-UNIT MAZE	20. Errors (2-4)
	21. Errors (13-15)
	22. Time (1-2)
	23. Time (6-12)
	24. Time (14-17)
	25. Starting Time
	26. *VTE*
6-UNIT DISCRIMINATION	27. Errors
	28. *VTE*
	29. Starting Time
	30. Time

Since the experiment had as its general aim the study of psychological differences resulting from genetic causes, it was considered of primary importance that no systematic environmental factors should be allowed to affect the strains differentially. Feeding, handling, living conditions, and exposure to the apparatus were therefore made as nearly equal as possible for all three groups. All of the rats were kept in a single rack of living cages, among which Brights, Dulls, and *MxM*'s were distributed systematically in order

to provide comparable physical and social living environments. Since single trials on some of the apparatuses required appreciable periods of time, it was impossible to make the presentation of problems exactly equal for all individual rats. It was also found impractical to randomize the order of running on each trial and on each piece of apparatus. Accordingly, each rat was assigned a rank at the beginning of the experiment and was run in that order, or its reverse, throughout. Constant differences among individual rats were introduced by this procedure, but since members of the strains were distributed, it served to maintain equal treatment of the groups and to facilitate the accurate recording of scores.

II. DESCRIPTIONS OF APPARATUS

A. Open Field

The Open Field apparatus for observing rats' emotionality responses was first described by Hall and Ballachey (5). As employed in the present experiment, it consisted of an 8-foot diameter circular area on the floor of the experimental room, enclosed by a sheet metal wall 12 inches high. The floor was marked off in numbered squares in order to facilitate making tracings on record sheets of the rats' paths within the area. Once a day for three consecutive days each rat was removed from his cage, placed in the center of the field, and allowed to remain there for a period of two minutes, during which time the measures listed below were taken:

1. *Defecations:* The total number of single faeces dropped.
2. *Urinations:* The number of separate urinations.
3. *Rearings:* The number of times the animal reared up on his hind legs.
4. *Washings:* The number of times the animal lifted forepaws to manipulate his face and whiskers.
5. *Distance:* Total distance represented by tracings of the rat's path, measured by chartometer.

B. Tryon Maze

The details of apparatus construction and procedures of running for the Tryon Maze have been described in a paper by Tolman, Tryon, and Jeffress (14). The standard procedure involves an initial 8-day period of preliminary training over a shortcut pathway (shown as a dashed line in Figure 3), followed by 19 days of running on the maze proper at one trial per day. The rats were given no food nor water throughout the entire period other than the reward which immediately followed each day's run, a soupy mixture of water and dry food. The present groups were run along with regular rats of the selective breeding experiment.

As an exploratory variation of the feeding procedure, two trials were added at the end of the 19th trial in which the rats were fed just before, instead of just after, the maze run. The automatic recordings of time and errors taken over all of the trials, 2 to 21, furnished data for the following variables:

6. *Errors:* Total number of full entrances into blind alleys during Trials 2 to 19, inclusive.
7. *Time:* Total number of time units, Trials 2 to 19.
8. *Errors:* Total number of blind alley entrances for Trials 20 and 21.
9. *Time:* Total number of time units, Trials 20 and 21.

C. Water Tank

The Water Tank apparatus was an 8-foot long galvanized iron tank, filled with water to a depth of 10 inches. The dimensions and essential parts are illustrated in Figure 1. The rats were placed by hand into the Starting Compartment, from which point they had to dive under the Starting Block and swim the length of the tank under water in order to escape. They were prevented from coming to the surface by a wire screen which extended nearly to the end of the tank, and were stimulated to remain close to the bottom by the tin "blocks" which hung from the screen at spaced intervals. The regions between these blocks constituted third-dimensional *culs-de-sac* which the rats learned gradually to eliminate over a long series of trials.

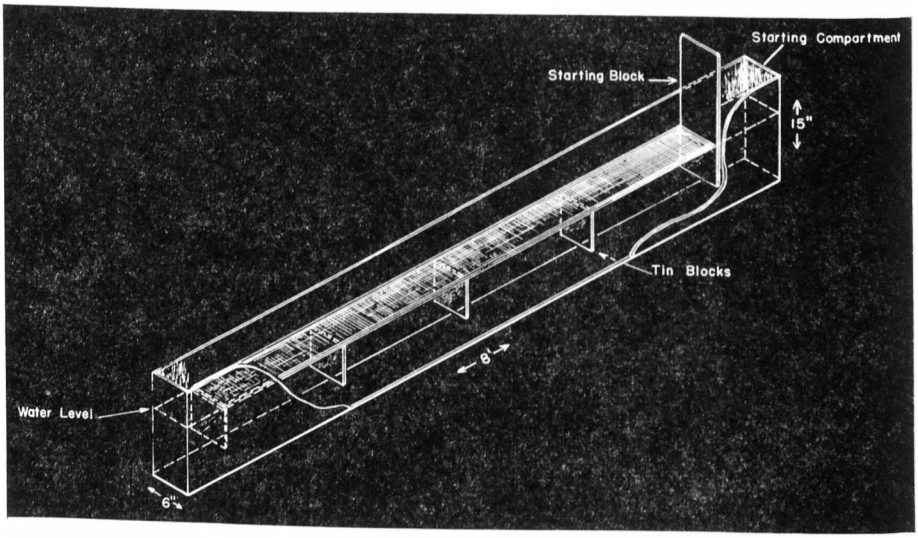


FIGURE 1
Water Tank

The procedure of running included a preliminary period during which the rats were trained gradually to swim under water. On the first trial they were placed into the water at the starting end and allowed to swim freely on the surface to a wire ladder leading out of the opposite end. Internal parts of the apparatus as shown in Figure 3 were suspended above the tank. Eight trials were given, during which the screen and blocks were lowered by successive stages into the water until, on the eighth trial, the screen was in its final position just under the surface. Eighteen further trials were then run, during which the following measures were obtained:

10. *Errors:* Each rat's score was taken as the sum of numerical values assigned to his entrances into the *culs-de-sac.* Full entrance up to the screen counted as "3," partial entrance as "2" or "1" depending on degree.

11. *Time:* Time was taken by stopwatch from the moment of diving under the surface to the moment of rising at the end.

12. *Starting Time:* After being placed into the Starting Compartment, the rats thrashed about or simply floated for some time before diving under water. These times were also recorded in seconds by stopwatch.

D. Water Unit

The Water Unit was a galvanized iron tank, similar in depth and width to the Water Tank described above but shaped in the form of a single *T*-maze unit. It contained water to a depth of 10 inches, and was so constructed that the rats could be made either to wade through or to swim through under water. Wading trials were made possible by a wire screen, placed four inches below the surface and extending to within six inches of the ends of stem, blind alley, and true path, with the remaining distance covered by tin gates (see Figure 2). For under-water trials these gates were raised on the stem and true path ends, opening the pathway under water and closing it off above. All ends of the tank were open above the 10-inch level, allowing the rats to enter and leave the apparatus just at the water surface. A starting compartment was attached at the stem end of the tank. This was a pan filled with water and covered over the top by a wire cloth cage, having a door in the rear for inserting the rat.

This apparatus was included as a problem in the present series because of the findings of a preliminary experiment in which Brights and Dulls had shown a marked difference in their readiness to dive voluntarily into the water. The procedure, which followed a preliminary training period on the Water Tank, involved giving the animals alternate wading and swimming trials in an insoluble right-left sequence, except that on each swimming (under water) run, the correct side was the same as on the immediately preceding wading run. Thus, a constant cue was provided which might have been learned, although none of the rats showed better than chance performance in this respect during the limited series of trials. The technique was to place each rat into the starting compartment by hand and allow him 60 seconds to enter the stem of the unit. If he had not entered at the end of that time he was given a light, intermittent sprinkling of water from above the compartment. Most rats would then dive off within a very few seconds. It was observed early in the experiment that on the first trial

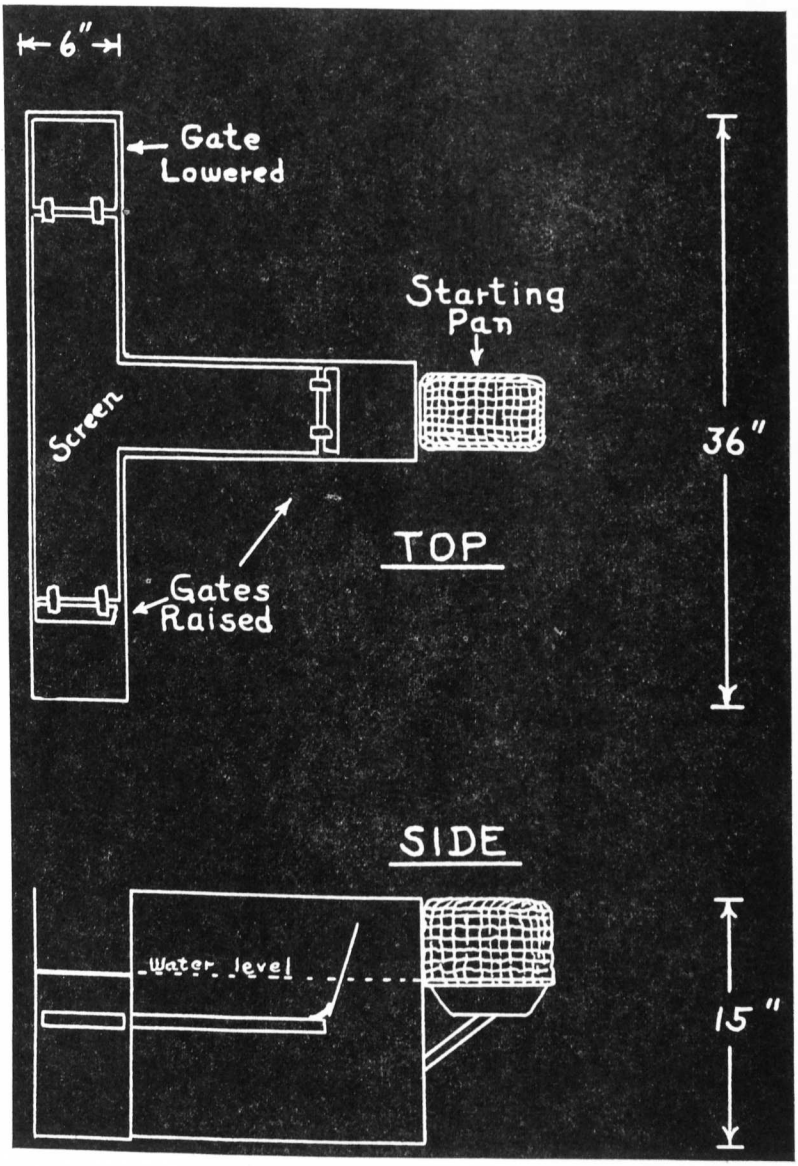


FIGURE 2
WATER UNIT

following their initial experience in making an under-water choice all members of the Dull group, without exception, left the starting compartment voluntarily, whereas the Brights, also without exception, remained the full 60 seconds and required the shower.

Essentially the same procedure was followed in the present experiment, and nearly identical results were obtained. Five wading and five under-water trials were given in alternate order.

13. *Starting Time:* The score of each rat was taken as the sum of starting times on under-water Trials 2 to 5, inclusive.

E. Elevated Path

The Elevated Path was 40 feet in length, two inches in width, and was mounted on standards to a height of 30 inches above the floor. It extended between two racks of sliding cages, one for starting the rats automatically and the other containing food as a reward at the end. A one-way gate was mounted on the path near the start and another near the end to prevent retracing.

During a series of preliminary trials the rats were accustomed gradually to the narrow pathway, gates, and end boxes. Following this period they were given eight trials at two trials per day. At the beginning of each trial they were placed in the sliding cage compartments at the start and were allowed to come out in turn to run the length of the path for food at the end. Times to run 40 feet between non-retrace gates were recorded by stopwatch.

14. *Time:* Each rat's score was the sum of his times in seconds for the eight trials.

F. Fourteen-Unit Maze

The pattern of the 14-Unit Elevated Maze is shown in Figure 3. Its pathway was two inches wide and 30 inches above the floor. An end-box rack, which is also shown in the diagram of Figure 3, had runners at two different levels for the end-boxes to slide on. The rats were first placed in individual compartments at the lower level, which connected to the pathway leading to the "Start" position. When the box was slid sideways for a distance of one compartment, a rat was freed to enter the maze. From the "End" position of the maze he entered a food compartment at the upper level. A total of 11 trials were given, during which the following measures were obtained:

15. *Errors:* Each rat's score was double the number of full and half errors during Trials 2 to 11, inclusive.

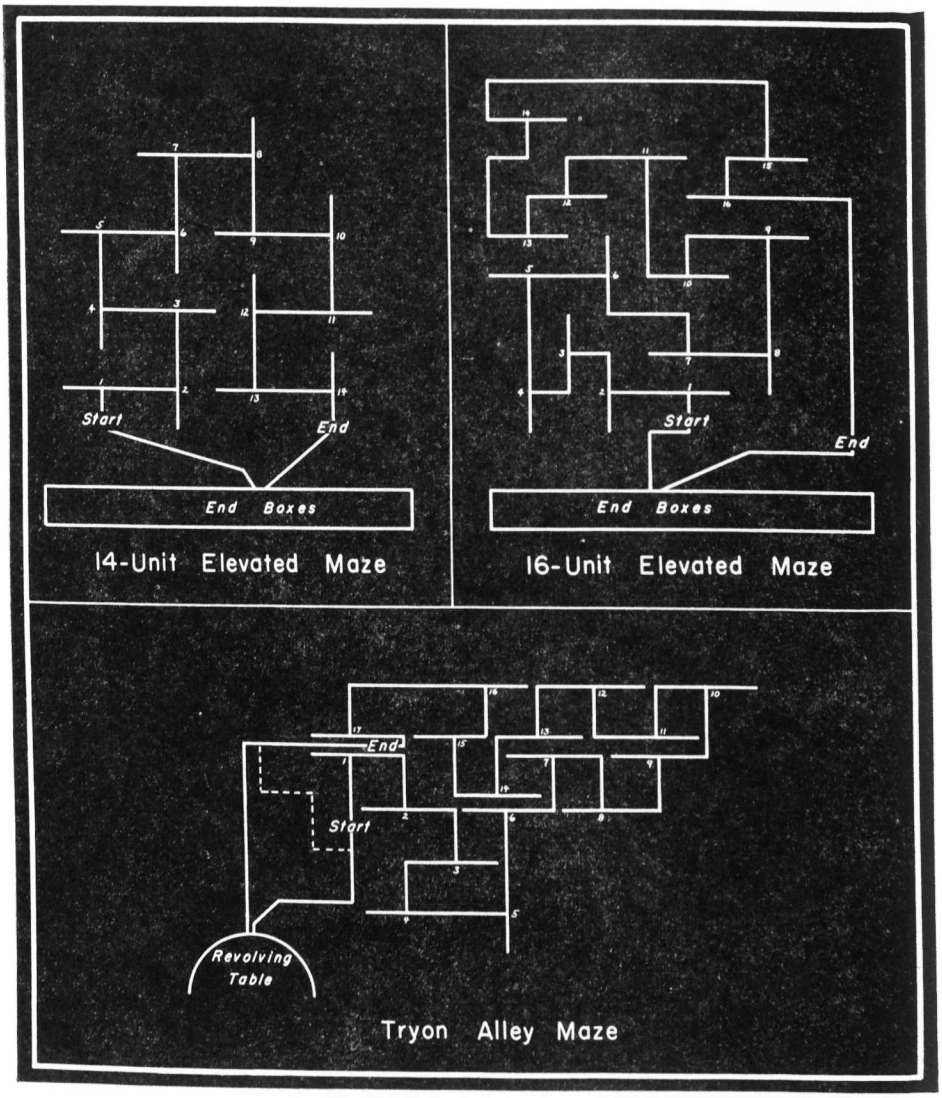


FIGURE 3
MAZE PATTERNS

16. *Time:* The total time in seconds from "Start" to "End," summed for the last 10 trials.

17. *Starting Time:* The time in seconds to come out of the starting compartment to "Start" position, summed for the last 10 trials.

G. Revolving Cages

The Revolving Cage apparatus was similar in general respects to others which have been commonly used for measuring rat activity levels. The wheels were mounted three tiers high on racks which stood against the walls of the experimental room. No stationary units were attached to the wheels. The details of their construction have been described in an earlier report (8).

The rats were not handled during the 15 days of running in the activity wheels. They were fed daily from 5:00 to 6:00 P.M., during which time the wheels were locked in a stationary position while their food dishes, containing wet mash, were left inside. The first eight days were allowed for "breaking-in," and thereafter counter readings were taken twice daily, at 3:00 and 5:00 P.M., for a period of seven days.

18. *Total Activity:* This measure was taken as the total number of revolutions for each rat during the seven days.

19. *Pre-Feeding Activity:* Scores were calculated for each rat as the ratio of activity during the two hours before feeding (i.e., 3:00 to 5:00 P.M.) to activity during the remaining 21-hour period of each day.

H. Sixteen-Unit Maze

The 16-Unit Elevated Maze was similar in construction to the 14-Unit Maze described above, but was of a more complex pattern. In Figure 3, patterns for both of these mazes and for the Tryon alley maze are shown for comparison. As in the case of the previous elevated maze, the rats started from boxes at the lower level of the end box stand and returned to food boxes at the upper level. Similar measures were taken of times and errors.

A total of 17 trials was given, at one trial per day. Error and time curves for all rats were found to plateau at about the 6th trial and remained relatively flat up to the 13th trial. At this point a variation in the room conditions was introduced in order to test the importance of visual cues in the rats' plateau performances. It had been noticed that the Dulls' error scores up to that time were very low, contrasting markedly with their Tryon Maze scores. Since visual factors are freer to operate in the elevated maze situation, this finding might have been expected on the basis of Krechevsky's theory that one of the fundamental differences between Brights and Dulls is a matter of

spatial and visual "hypotheses." If it should be found that Dulls were more disrupted than Brights by the *removal* of visual cues, additional support would be lent to Krechevsky's theory. Accordingly, it was decided to run a series of extra trials with the experimental room blacked out.[5]

Following the 12th trial, the room was completely sealed against external light and the rats were run for the remaining trials under two conditions of darkness: (*a*) On Trials 13 to 16, inclusive, a spotlight equipped with a red filter was used by the experimenter in order to follow the rats' paths around the maze and record errors.[6] (*b*) On Trial 17, an automatic switch was employed for recording total times and the rats were run in complete darkness. Both errors and times were recorded on Trials 13 to 15, and time only on Trials 16 and 17.

On the basis of learning curves for time and errors, the 17 trials were divided into initial, plateau, and test trials, and scores were summated separately for these three periods. The following measures were used:

20. *Errors:* Double the number of full plus half errors during Trials 2 to 4, inclusive.
21. *Errors:* Errors during Trials 13 to 15, inclusive.
22. *Time:* Total of times for Trials 1 and 2.
23. *Time:* Total of times for Trials 6 to 12, inclusive.
24. *Time:* Total of times for Trials 14 to 17, inclusive.
25. *Starting Time:* Total of starting times for Trials 14 and 15.
26. *VTE:* "Vicarious trials and errors" were counted as the number of times the rat made sideways head movements at the choice point before taking one or the other pathway. Totals were taken for Trials 5 to 15, inclusive.

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[5]The failure of disruption to occur would not, however, disprove the hypothesis, since plateau performances might be relatively independent of the specific cues used in learning. Tryon (17, 19) has shown that complete and sudden removal of visual cues during the plateau period of learning the *alley* maze has no effect on individual differences in errors and only a very slight effect on speed of running. Nevertheless, it is possible that on the elevated maze, where visual cues are more prominent, both the learning and plateau performances depend to a greater extent upon visual control. If so, differential sensory preferences might have a greater effect on the relative scores of Brights and Dulls in this apparatus.

[6]A light of low intensity was used with the filter, which transmitted essentially no light below 580 millimicrons. Although no conclusive proof exists that rats are unable to see light above this point, their behavior during the experiment appeared to justify assuming that they could not. In Figure 6, where the total time curves have been reproduced, it is evident that a marked disruption occurs with introduction of the red light at Trial 13, whereas no change can be detected at Trial 17 between the red light condition and total darkness.

I. Six-Unit Discrimination

This apparatus consisted of six consecutive discrimination units, each of which was essentially a modified Lashley jumping stand containing two possible openings for the rat to pass through. The openings were covered by light-weight gates, hinged at the tops and fastened (in the case of the "wrong" gates) at the bottoms. White, black, or neutral color cards were pasted on the fronts of these. Passing from one unit to the next, the rat jumped a gap of six inches to a narrow platform in front of the two gates. A vertical partition separated the gates, forcing the rat to jump back across the gap when he had chosen the wrong side. The platform in back of each unit was narrowed towards the center, making it necessary for the next choice to be made from a central position.

The rats were started from sliding cages (rather than being placed on the initial jumping platform by hand) and were run through the six units to sliding cages containing food at the end. Fourteen preliminary trials were given for training in jumping and in operating the gates, during which neutral gray cards were pasted to all gates and the right-left sequence of correct sides was randomly varied.

The test problem was set up as an ordinary black-white discrimination-learning series, except that the right-left positions of black and white gates were not varied. The white-correct gates were arranged in the order, *L-R-R-L-L-R*, and were kept in that order throughout the series. This made it possible for the problem to be learned either in terms of the constant visual cue or as a constant spatial-temporal pattern.

The rats were run two trials per day for 24 trials, at the end of which all groups had reached an average level approaching zero errors. This was the learning series, during which the following measures were taken for inclusion in the 30 test variables:

27. *Errors:* A maximum of one error could be made in each unit; scores were the total errors for 24 trials.
28. *VTE:* One "vicarious trial" was counted for each head movement in the direction of either gate.
29. *Starting Time:* Time in seconds spent on the starting platform before jumping to the first unit.
30. *Time:* The remaining time in seconds up to the point of reaching the end cages.

On the 25th trial, the right-left sequence of white-correct gates was reversed to the new order, *R-L-L-R-R-L*, as an exploratory comparison of Brights and Dulls with respect to relative strengths of the spatial and visual

cues. Four trials, including the 25th, were run under these conditions. The results of this test may be mentioned here briefly as of some possible interest for future study, but they were somewhat ambiguous and are not sufficiently reliable to be included with the other data. The Brights' errors increased nearly to "chance" on the 25th trial, whereas Dulls were only slightly disrupted. On the surface, this result is similar to Krechevsky's previous finding with regard to spatial and visual "hypotheses," but the other measures which were taken at the same time indicate that in this case a somewhat different principle may be involved: (*a*) As indicated by their starting times and *VTE*'s[7] both the Brights and Dulls were noticeably disrupted at the *first* unit, while (*b*) Brights' errors were reduced immediately on the following trials. These observations indicate that both groups learned both cues, and that the differentiation here is therefore one of *preference* for spatial or visual—after learning has occurred—rather than of contrast in processes which contribute to the rats' learning. Thus, the present results provide no direct evidence either for or against the theory that a "spatial" tendency underlies the Brights' superior alley maze performances. The experiment should be repeated with larger groups and with different types of variations being introduced at the crucial trial.

[7] *VTE*'s (vicarious trials and errors) are ordinarily considered to be a "cognitive" rather than an emotional behavior (cf. Ref. 4, p. 19). But in the case of this particular trial on the discrimination apparatus, the increase in back-and-forth head wags was clearly a sign of disruption.

III. STATISTICAL ANALYSIS

A. Standardization of Test Measures

The raw scores for all rats in each of the 30 tests are listed in Table 3. To convert these into units of a common scale, using *MxM*'s as the standardization group, a table was prepared (Table 4) by means of which the *MxM* raw scores in each variable, when ranked, could be converted directly to normalized values on a scale ranging from 0 to 100. In constructing the table, ranks were first transformed into percentile ranks and then into their sigma score equivalents as found in a table of normal curve areas. The sigma scores were then converted to final scale values by means of the formula, $S = 20z + 50$.

The final scale had, therefore, a mean of 50 and a standard deviation of 20. Because of the small number of rats in the *MxM* group, their scores actually covered only the range between 13 and 87 (corresponding to —1.83 and +1.83 sigma units, respectively), the full scale from 0 to 100 having been allowed in order to cover a few instances in which individual Brights' and Dulls' raw scores were definitely above or below those of the extreme *MxM*'s. Ideally, the limiting scores would be set farther out on the scale in terms of sigma units, since it is reasonable to expect that members of the pure strains might deviate as much as three standard deviations in some variables. By the same token, a much larger sample of *MxM*'s would be highly desirable, both to eliminate the need for extrapolation and to increase the reliability of obtained points within the distributions. It was found, however, that the majority of Brights and Dulls were within the *MxM* ranges on most of the tests and it was therefore considered inadvisable to extend the limits beyond those chosen.[8]

In order to transform individual Brights' and Dulls' raw scores into the new scale values, a graph was made for each variable on which *MxM*'s normalized scores were plotted as a function of their raw scores. A smooth curve was drawn through the plotted points and scores for the experimental groups were read directly from these graphs. Since approximations were necessarily involved in the smoothing process, the reading of normalized scale values could not be considered highly accurate. Hence, the Brights' and Dulls' scores were read only to the nearest multiple of five units. The results are listed in Table 5.

[8]Unreliability of the scales caused by random variation in means and sigmas of the small *MxM* sample obviously does have some effect upon the obtained patterns of scores for Brights and Dulls, but the nature of the effect is difficult to estimate. Considering the probable amounts of sampling error, the influence upon relative values of the correlations within and between strains should be negligible.

TABLE 3

TABLE OF RAW SCORES FOR BRIGHTS, DULLS, AND *MxM*'S IN THE 30 TEST VARIABLES

Rat	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
G- 6	16	1	4	0	10	16	209	5	93	60	134	198	267	115	20	160	76	2212	111	36	4	565	300	353	13	21	51	218	5	12
G- 5	14	2	7	4	24	47	399	11	70	53	130	285	255	150	21	150	124	7482	141	20	2	413	298	216	18	20	40	255	7	18
G-10	8	1	11	3	114	27	224	17	98	63	130	48	287	90	14	140	74	2737	28	18	0	516	226	175	14	6	27	232	4	16
G-12	10	7	27	23	69	16	179	7	56	132	191	112	270	140	21	165	87	11301	138	8	10	468	283	178	11	9	29	230	3	13
G-16	15	7	24	4	89	23	208	2	40	72	153	34	301	115	9	175	87	762	153	26	25	651	404	370	17	19	25	279	4	12
G-18	17	7	31	10	100	20	162	2	30	72	129	31	236	115	19	160	83	4974	72	7	10	530	411	272	14	28	38	220	3	23
G-24	8	1	7	0	55	17	160	9	44	73	133	30	625	135	15	115	74	3672	195	11	20	440	341	303	16	17	30	203	7	11
G-27	9	2	14	31	67	16	146	14	71	75	126	26	181	105	12	150	82	3539	207	24	15	540	279	272	15	21	42	208	6	24
G-28	4	1	23	13	76	14	145	3	27	35	122	172	247	125	11	145	69	5881	449	25	10	346	272	189	12	12	44	211	2	11
G-30	8	0	17	23	112	28	175	10	58	86	140	34	302	115	10	155	145	2358	96	18	1	416	290	209	24	3	39	275	23	44
B- 3	0	1	34	0	136	135	540	12	102	38	108	30	139	155	17	150	52	33013	39	14	2	387	325	344	24	15	31	306	22	34
B- 6	1	0	35	4	140	124	656	14	212	28	102	21	115	240	23	290	181	5755	47	43	7	345	571	411	47	38	21	400	30	60
B- 8	0	1	32	3	185	98	480	17	142	39	109	112	134	220	26	175	80	27623	52	22	7	485	443	391	19	20	39	297	30	55
B-10	0	0	32	14	139	122	456	14	94	21	103	51	118	210	24	235	86	10580	105	24	6	415	302	255	27	11	46	275	12	34
B-12	0	1	22	5	77	124	820	15	160	27	100	35	147	150	20	170	74	30485	44	54	5	549	353	368	17	37	31	244	6	19
B-15	7	0	9	4	26	89	810	9	177	38	97	36	113	160	17	220	190	6560	133	24	5	503	464	392	45	50	19	294	20	26
B-18	0	0	31	3	97	149	517	16	97	11	86	33	126	135	26	250	161	13329	55	19	4	421	543	373	32	30	36	361	32	26
B-22	0	0	26	14	96	120	396	16	103	31	107	26	164	170	13	155	129	19438	81	39	11	390	353	296	35	18	46	243	8	34
B-24	14	0	21	23	126	147	698	19	140	57	108	59	144	160	17	130	109	10315	63	16	4	308	281	346	24	9	60	194	8	24
B-26	0	2	49	3	164	135	594	18	176	14	111	26	152	255	27	265	170	8122	59	14	4	338	348	390	23	11	23	295	12	27
BH 5	17	9	5	2	14	108	390	10	48	19	90	26	180	120	19	145	110	20533	93	23	5	370	283	197	81	22	48	245	6	27
BH 9	5	4	21	4	137	43	175	10	51	52	107	40	193	105	13	135	78	50629	38	44	28	344	314	253	18	30	32	198	3	8
BH11	8	1	15	10	78	42	286	10	54	19	116	27	154	125	21	175	111	15589	74	32	32	362	329	308	21	23	46	264	7	16
BH13	2	3	23	2	159	74	330	20	180	20	109	20	153	150	6	140	150	16000	63	45	9	379	400	231	20	31	27	306	12	20
BH17	0	1	25	4	199	32	268	10	66	19	97	35	217	130	28	160	145	36693	41	30	20	526	353	266	9	33	43	221	4	11
BH22	2	6	25	14	188	132	334	11	74	46	124	50	199	120	25	225	114	25272	35	24	22	361	301	329	24	22	40	230	6	24
BH26	0	1	19	8	214	32	152	6	34	35	95	18	134	100	16	140	160	5881	260	38	5	437	302	192	18	6	38	240	14	22
BH30	0	5	24	12	181	33	192	16	103	30	120	21	99	95	7	125	124	44736	55	34	14	366	333	227	23	34	38	197	3	13
BH33	0	0	29	0	172	56	231	10	59	20	122	153	225	105	22	110	52	2747	109	32	28	401	265	334	13	13	59	230	9	27
BH37	0	2	21	6	200	55	279	19	182	57	100	19	186	135	11	130	88	10761	118	24	2	272	277	290	31	12	35	210	3	11
BH41	10	10	34	18	70	55	274	8	82	63	106	23	169	210	18	180	137	8066	53	38	18	347	398	400	35	28	42	237	6	19
BH45	0	1	11	7	170	102	350	17	77	27	99	41	186	155	22	135	127	25839	55	10	17	389	371	324	47	24	39	272	13	36
BH48	0	1	35	2	192	39	189	17	103	23	127	34	179	100	17	115	64	23074	62	49	21	407	275	177	21	11	37	200	6	14
BH54	4	2	56	8	133	60	240	14	65	60	110	72	192	115	23	150	99	14752	68	32	25	433	273	238	18	5	47	208	6	21
BH56	7	2	14	14	146	44	212	10	47	49	105	66	127	110	14	125	99	18959	46	39	19	399	221	173	13	10	41	175	2	10

TABLE 4

TABLE FOR CONVERTING RANKED SCORES OF THE *MxM* GROUP INTO FINAL SCALE VALUES

Rank	% Rank	*z*	*S*
1	96.67	1.83	87
2	90.00	1.28	76
3	83.33	0.97	69
4	76.67	0.73	65
5	70.00	0.53	61
6	63.33	0.34	57
7	56.67	0.17	53
8	50.00	0.00	50
9	43.33	—0.17	47
10	36.67	—0.34	43
11	30.00	—0.53	39
12	23.33	—0.73	35
13	16.67	—0.97	31
14	10.00	—1.28	24
15	3.33	—1.83	13

B. DISTRIBUTIONS OF SCORES FOR INDIVIDUAL TESTS

The data of Table 5 are presented graphically in Figure 4, which shows histograms drawn for each of the 30 test measures. The baseline of each figure represents the possible range of values from 0 to 100, extending from left to right. Individual Bright rats are represented as crosses above the line and the Dulls as circles below.

Because of the method by which the scores were obtained, it is to be assumed that *MxM* rats distribute normally on each of the 30 test measures, having means of 50 throughout. This background comparison is taken to be implicit in all of the Brights' and Dulls' scaled scores as they are discussed in the following sections.

It may be noticed in comparing Table 5 with the histograms that the direction of "high" and "low" has been reversed in many cases from that of the original scores. For example, in the case of Variable 6 (errors in the Tryon Maze), a high scale value represents a low number of errors, and in Variable 7, the high scale value represents a low total time score. All time and error scores were treated in this way in conformity with the accepted notion of what constitutes "good" and "poor" learning performances. Some of the other variables, however—e.g., Variable 19 (percentage pre-feeding activity) and Variable 28 (*VTE*'s in discrimination learning)—required an arbitrary decision. In such cases, the direction chosen to be "high" was that which made for positive correlation within the *MxM* group with other measures taken in the given apparatus. Since time and error scores constituted

TABLE 5

TABLE OF SCALED SCORES FOR BRIGHTS AND DULLS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Rat #
15	65	15	85	15	100	60	85	35	25	10	10	5	55	45	G- 6
20	50	20	55	20	50	15	60	50	35	10	0	5	30	40	G- 5
30	65	25	60	35	100	55	30	35	15	10	35	5	80	60	G-10
25	30	60	10	25	100	70	75	60	0	0	20	5	35	40	G-12
15	30	55	55	30	100	60	95	80	5	5	50	0	55	75	G-16
15	30	65	40	30	100	85	95	90	5	10	50	10	55	45	G-18
30	65	20	85	25	100	85	70	75	5	10	50	0	35	60	G-24
30	50	30	0	25	100	90	45	50	5	15	55	45	65	65	G-27
40	65	50	30	25	100	90	95	90	45	25	10	10	45	70	G-28
30	85	40	10	35	100	75	65	55	5	10	50	0	55	70	G-30
65	65	70	85	40	15	0	55	35	45	50	50	70	25	50	B- 3
55	85	70	55	40	20	0	45	10	50	60	65	80	5	35	B- 6
65	65	70	60	55	25	0	30	25	45	45	20	70	10	20	B- 8
65	85	70	30	40	20	0	45	35	60	55	35	80	15	30	B-10
65	65	50	55	30	20	0	40	20	55	60	45	65	30	45	B-12
35	85	25	55	20	30	0	70	20	45	70	45	80	25	50	B-15
65	85	65	60	30	10	0	35	35	95	90	50	75	35	20	B-18
65	85	60	30	30	20	10	35	35	50	50	55	60	20	65	B-22
20	85	45	10	35	10	0	25	25	30	50	30	65	25	50	B-24
65	50	80	60	50	15	0	25	20	90	45	55	65	0	20	B-26

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	Rat #
35	70	10	30	45	75	10	50	25	75	50	25	45	55	65	G- 6
40	45	30	20	75	80	35	50	60	60	50	50	65	40	50	G- 5
50	70	15	90	75	90	15	85	80	70	75	85	55	60	55	G-10
35	65	35	20	90	65	20	60	75	80	70	85	55	70	60	G-12
30	65	5	15	65	35	5	20	20	60	50	90	75	60	65	G-16
35	65	20	40	90	65	10	15	45	70	45	55	45	70	40	G-18
75	70	15	10	85	45	25	40	40	65	55	80	35	40	65	G-24
40	65	15	5	70	60	10	60	45	70	50	45	40	45	35	G-27
45	75	25	0	65	65	70	65	70	75	60	40	40	85	65	G-28
40	30	10	35	75	90	35	55	65	40	90	55	75	10	10	G-30
40	85	65	70	80	80	45	45	25	40	55	80	85	10	15	B- 3
0	5	25	60	30	70	70	0	10	25	10	95	100	0	0	B- 6
30	70	65	60	70	70	20	10	15	55	50	55	85	0	5	B- 8
10	60	35	30	70	70	35	50	50	40	65	35	75	30	15	B-10
30	70	65	65	10	75	10	35	20	60	10	80	60	45	50	B-12
15	5	25	20	70	75	15	10	15	25	0	100	85	10	30	B-15
10	10	40	55	75	75	30	0	20	35	40	65	95	0	30	B-18
40	40	50	40	40	65	45	35	40	30	55	35	60	35	15	B-22
60	50	35	50	80	75	80	60	25	40	70	10	30	35	35	B-24
5	10	30	50	80	75	75	35	15	45	65	95	85	30	30	B-26

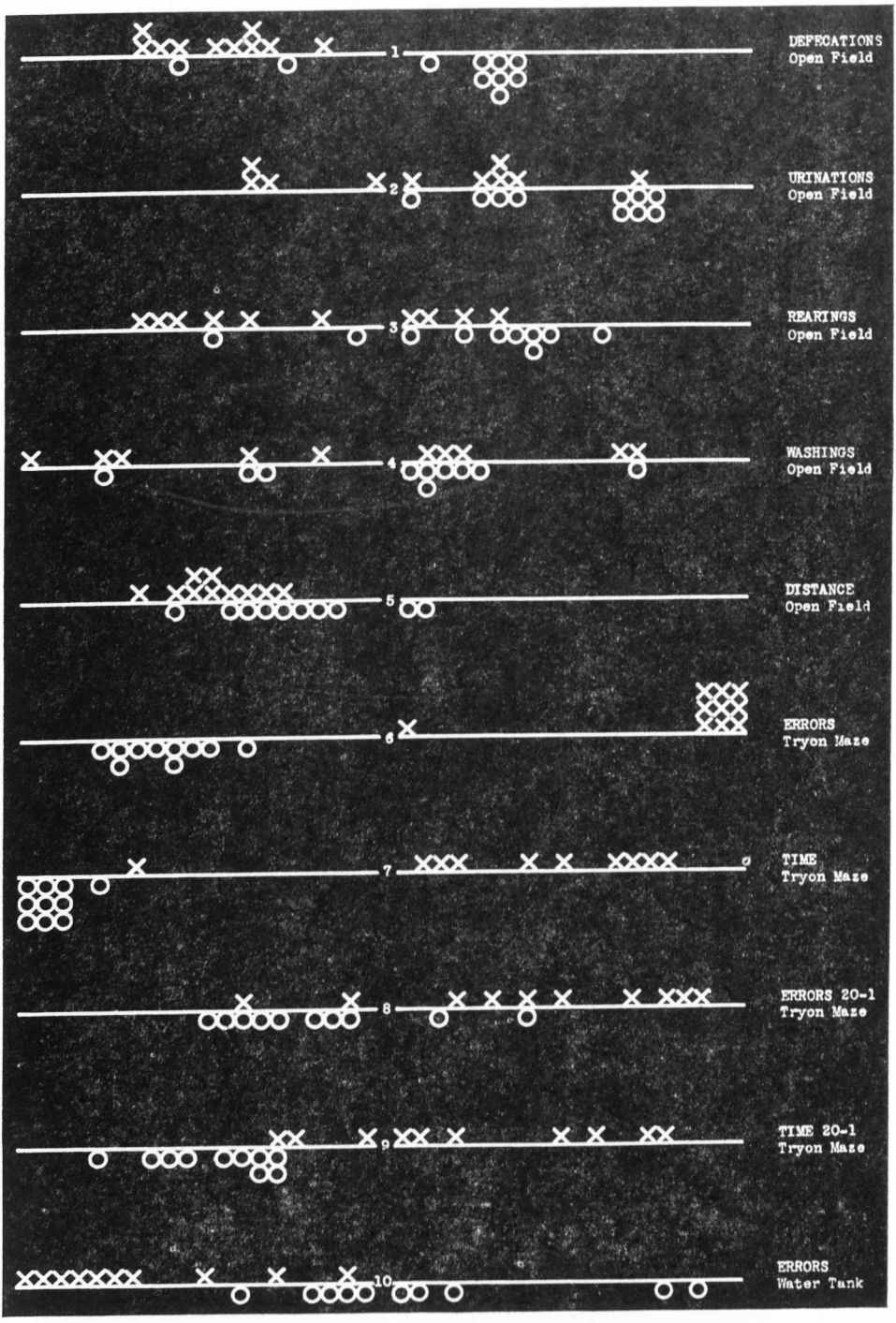


FIGURE 4*a*

HISTOGRAMS OF INDIVIDUAL VARIABLES: VARIABLES 1-10

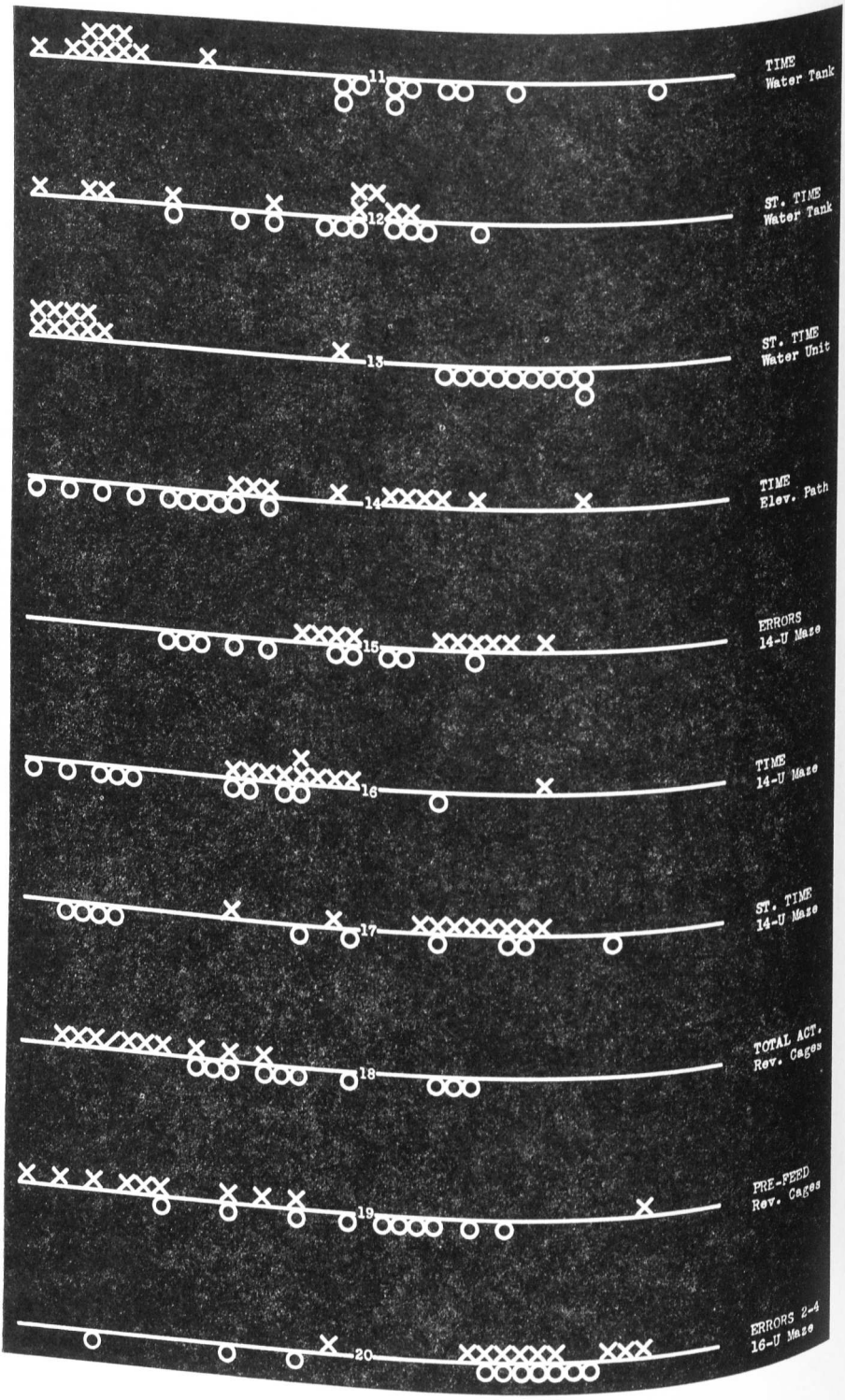


FIGURE 4*b*

HISTOGRAMS OF INDIVIDUAL VARIABLES: VARIABLES 11-20

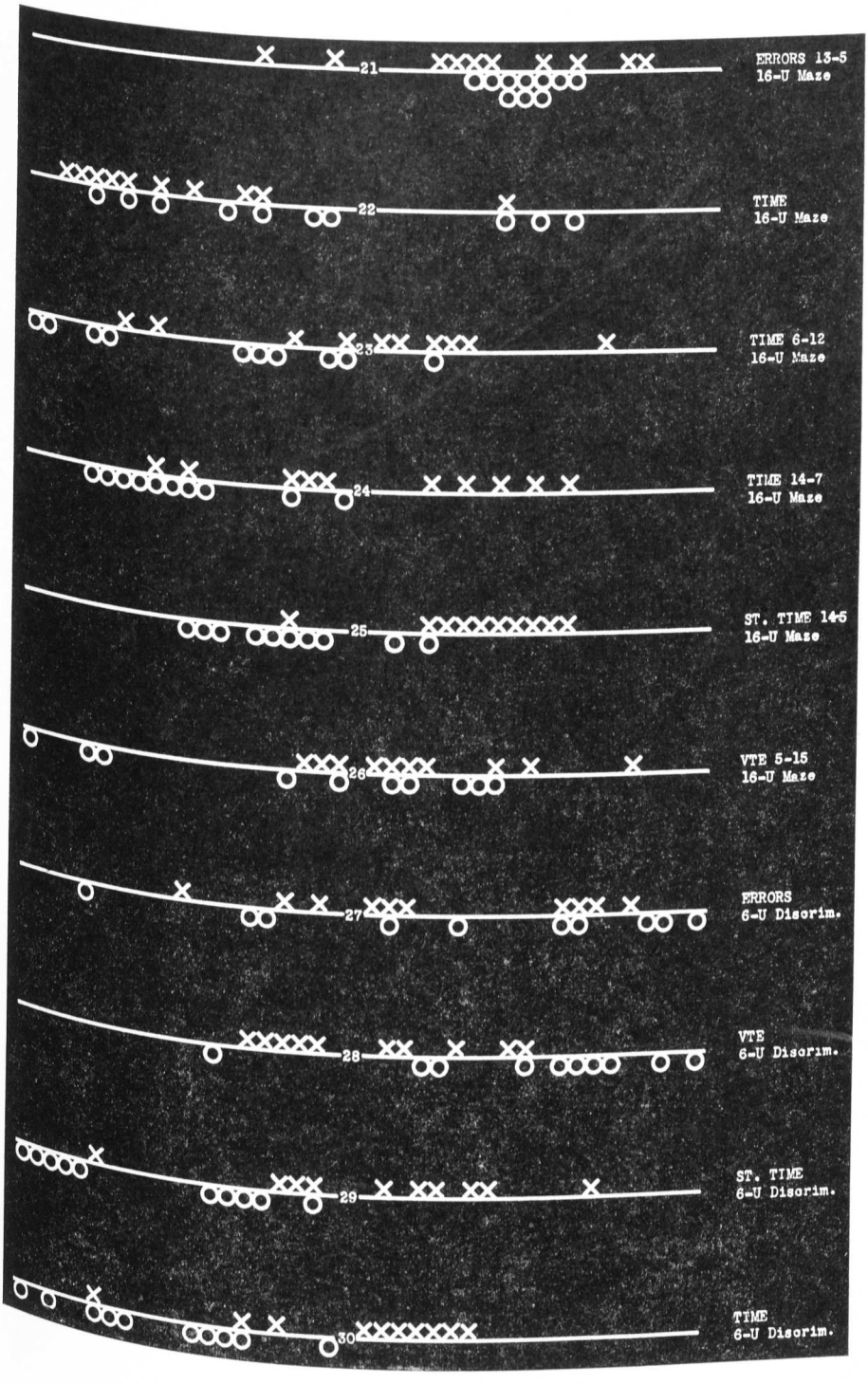


FIGURE 4*c*

HISTOGRAMS OF INDIVIDUAL VARIABLES: VARIABLES 21-30

most of the measures taken, only four of the 30 test variables were *not* reflected in the process of scaling. These non-reflected variables are listed below:

3. *Rearings:* Open Field.
5. *Distance:* Open Field.
18. *Total Activity:* Revolving Cages.
28. *VTE:* 6-Unit Discrimination.

C. Intercorrelations Among Rats

It is generally true of all methods of multiple factor analysis that the variables which have high loadings in any given factor (i.e., which form a cluster in common factor space) are variables which correlate highly with one another and low with other variables outside the cluster. Hence, if it can be shown in the present case that individual Bright rats correlate highly with one another and low with individual Dulls, and that Dulls also correlate highly among themselves, it may be assumed that Brights and Dulls would form two separate clusters in common factor space if a complete factor analysis were performed on the data. Since the question of principal interest in the present experiment involves the possible existence of only these two types, and since members of the types are known in advance, it was considered economical to test the hypothesis directly by simply finding the average intercorrelations within and between groups.

Accordingly, averages were computed for (*a*) the correlations of Bright rats with other Brights, (*b*) of Dulls with other Dulls, and (*c*) of Brights with Dulls. In order to provide a means of judging the representativeness of group averages, mean correlations were also computed separately for each rat.[9] The results are shown in Table 6.

With reference to the hypothesis that Brights and Dulls are different behavior "types," it may be seen by referring to Table 6 that the obtained correlations definitely indicate a positive answer. The fact that the negative correlation *between* groups (—.19) is much smaller in magnitude than the average positive correlation *within* each group (+.56) indicates, further, that the two strains are not merely opposites but are comparatively unrelated. This finding is inconsistent with the conventional assumption of simple, quantitative differences among individuals: it implies that different *patterns* of components must be postulated to underlie maze-brightness and maze-dullness.

[9] The 190 possible intercorrelations among individual rats were not computed. Instead, the values listed in Table 6 were obtained by means of formulae for average intercorrelations, based upon scores which were first standardized for each rat with reference to his distribution of scaled scores in the N test variables. No approximations were necessary; the obtained values represent true arithmetic means of the intercorrelations.

TABLE 6
AVERAGE INTERCORRELATIONS

Brights	Mean r with Brights	Mean r with Dulls
*A. Averages for Individual Rats**		
G- 6	.592	—.198
G- 5	.559	.079
G-10	.522	—.152
G-12	.663	—.233
G-16	.606	—.195
G-18	.629	—.260
G-24	.624	—.268
G-28	.568	—.353
G-30	.540	—.025
Mean Bright	.587	—.188
Dulls		
B- 3	—.140	.609
B- 6	—.270	.602
B- 8	—.155	.597
B-10	—.136	.591
B-12	—.244	.431
B-15	—.016	.502
B-18	—.300	.619
B-22	—.265	.555
B-24	—.112	.273
B-26	—.238	.554
Mean Dulls	—.188	.533

B.—Averages for the Groups

Group	No of rats	No. of inter-r's	Ave. inter-r
Bright vs. Bright	10	45	.587
Dull vs. Dull	10	45	.533
Bright vs. Dull	20	100	—.188

*The mean r of each rat with his own group represents the average of nine correlations; that with the opposite group, the average of 10.

Since satisfactory formulae were not available for computing standard errors of average intercorrelations, precise estimates could not be made of their statistical significance. In view of the non-random character of the sampling, standard errors which assume simple sampling would probably be too high. That is, since the Bright and Dull groups had been selected by taking one member from each litter of the given generation, they were probably more representative of their respective populations than estimates based on conditions of random sampling would indicate. Nevertheless, a fair evaluation of the reliability of group averages may be made by consulting the values in Table 6 for individual rats. Whereas the correlations of

individual Brights with other members of their own group range from +.52 to +.66, their correlations with members of the Dull group range from —.35 to +.08. Similarly, the individual Dulls correlate from +.27 to +.62 with other Dulls, but from —.30 to —.02 with Brights. In comparison with the differences between groups, the ranges of intra-group variation are very small. Since these figures indicate that members of the two strains can be distinguished with perfect accuracy on the basis of their score patterns alone, there can be little question that the group differences are significant statistically.

In evaluating the figures of Table 6 it should be kept in mind that the obtained values were subject to influence by both (*a*) unreliability of the test measures and (*b*) variable factors of previous training and experience. As regards the problem of the extent to which innate determiners are responsible for differences between Brights and Dulls in their trait organization, both of these secondary types of influence are spurious, and both would operate in such a way as to lower the obtained correlations. If, therefore, the "true" values could be computed, it is reasonable to suppose that they would show both a closer similarity of members within each strain and a sharper difference between the strains.[10]

[10]Estimates might be made of these true values by either of two methods, neither of which was used for the present report because its value did not seem to justify the required labor to compute: (*a*) Attenuation by factors of measurement could be corrected for if reliability coefficients were first computed for each rat. The number of these coefficients, 20, is not great, but to compute them would also require the preliminary scaling of 60 split-half score distributions. (*b*) If a complete inverted-factor analysis were made, the relationships when expressed in terms of "common factor" correlations should approximate the true degrees of hereditary resemblance, since the effects of both unreliability and specificity (which, presumably, would be increased by environmentally-produced differences among rats) would be removed.

IV. PSYCHOLOGICAL ANALYSIS OF BEHAVIOR PATTERNS

A. Graph of Bright and Dull Medians

In order to obtain a simplified picture of the Brights' and Dulls' performances on all of the variables, a chart was prepared which is reproduced in Figure 5. The chart is constructed in the form of a scatter diagram, showing the position of each test measure with respect to the median scores of the two groups of rats. Scale units of the ordinate, representing Dulls, and of the abscissa, representing Brights, are in terms of the final scaled scores, which range from 0 to 100. Each test measure is plotted as a point whose coördinates are the medium scores of Brights and of Dulls, respectively, in that test. Thus, for example, the coördinates of Variable 6 are 100 for Brights and 25 for Dulls, these two values having been determined as the respective medians in that variable (cf. histograms of Variable 6 in Figure 4).

In general, since the score of 50 indicates the constant position of the median *MxM*, all points in the lower right hand quadrant are variables in which the median Bright is above, and the median Dull below, the average *MxM*. The reverse is true for points in Quadrant II (upper left), while in Quadrants I (upper right) and III (lower left) both groups are above and below average, respectively.

The scattering of points indicates a slight negative correlation between the median Bright and median Dull, as might have been expected in view of the previously obtained mean correlation of —.19 between individual members of the two groups.

The succeeding analysis of group differences is based upon these median profiles. As a measure of the extent to which they may be considered representative of the profiles of individual rats, each rat's score distribution was correlated with the median for his group. It was found that Brights correlated +.78, on the average, with the median Bright profile and Dulls +.76 with the median Dull profile. Although far from indicating perfect homogeneity, these figures are sufficiently high to justify accepting the median profiles as characteristic of their respective groups.

B. Generality of Brightness and Dullness

By analyzing the scatter of variables in Figure 5 it is possible to note the degree to which obtained scores are consistent with various hypotheses concerning the nature of specific behavior differences.

A consideration of primary interest concerns the extent to which there is a general superiority of Brights over Dulls. This does appear to be the

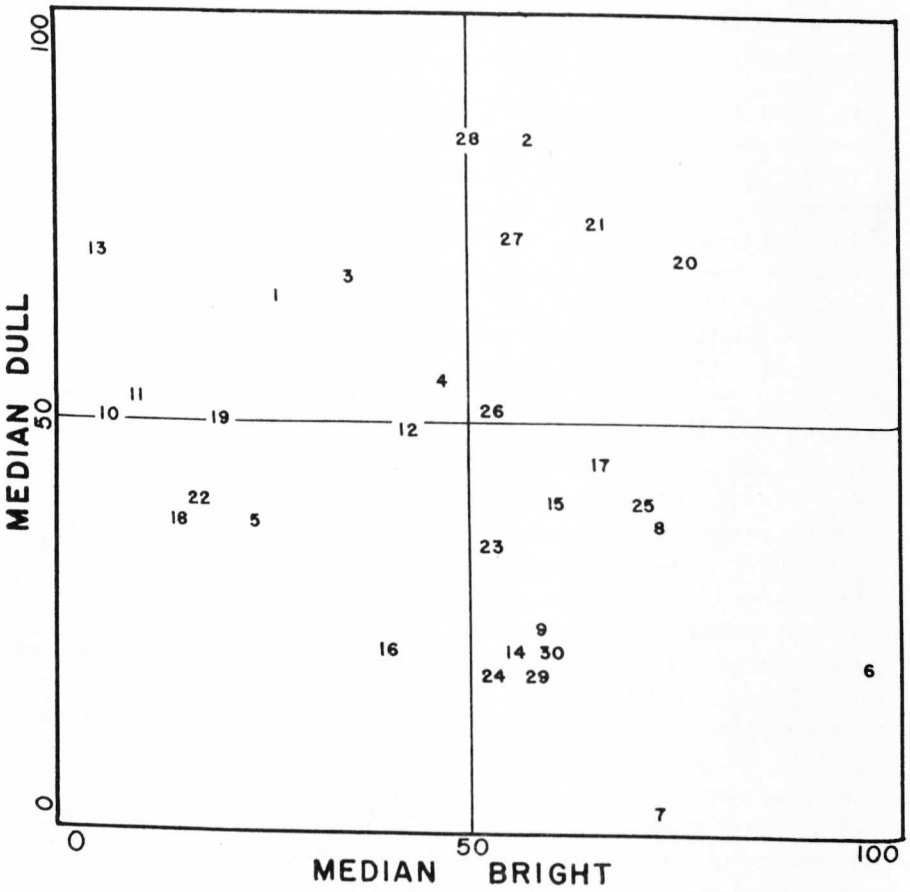


FIGURE 5

Scatter diagram showing the relative positions of Bright and Dull medians in all variables. Each numbered point in the plot represents a test variable, the locus of which is determined, on the abscissa, by the median score for Brights on that variable, and on the ordinate by the median score for Dulls.

case to a slight degree, as it may be observed that the variables lie to the right of the center vertical line, and below the center horizontal line, in more than half of the cases, indicating that Brights are "high" and Dulls are "low" in a majority of the measures. The difference, however, is too small to be considered reliable. It is significant that Dulls were actually superior to *MxM*'s in 13 of the measures (i.e., Variables 1, 2, 3, 4, 10, 11, 13, 19, 20, 21, 26, 27, 28) while Brights were below average in 12 (Variables 1, 3,

4, 5, 10, 11, 12, 13, 16, 18, 19, 22), a finding which obviously indicates that Dulls are not *universally* inferior rats and that Brights are not universally superior.

Of particular interest however, are the measures of learning. Variables 6 (Tryon Maze), 10 (Water Tank), 15 (14-Unit Maze), 20 (16-Unit Maze), and 27 (6-Unit Discrimination), all being measures of error, are the five measures which best represent the rats' learning performances under a variety of conditions. Referring to the positions of these variables in Figure 5, it may be seen that the Brights are above average in 6, 15, 20, and 27, and very low in 10, while the Dulls are low in 6 and 15, average in 10, and definitely superior in 20 and 27. The Dulls, then, are either equal to or better than Brights in three of these five measures. In view of this result, it is apparent that the performances of the two groups on any particular apparatus are not to be accounted for in terms of a difference in general learning ability.

C. Learning in Food-Reward Situations

Of the five learning variables discussed above, the median Bright score was found to be below average only in Variable 10. This variable is also the only one of the series in which the reward employed was escape-from-water rather than food. When only those measures representing errors in food-reward situations are considered, it is found that the Brights made low error scores consistently. Thus in Figure 5*a*, where the food-reward error measures are graphed separately,[11] all of the points appear to the right of the center vertical line, indicating that the Brights were generally superior in eliminating blind alley entrances when the incentive was food. On the basis of this finding, one is tempted to suggest that the error scores are determined primarily by degrees of food motivation. However, scores for the median Dull are distributed throughout both Quadrants I and IV, and such an interpretation, if applied to all three groups of rats, would imply that Dulls were approximately equal to *MxM*'s in food drive. In view of the previous indications that the groups are differently organized, the possibility is still not precluded that Brights' superiority is due to the strength of their food drive. However, conclusions regarding this point should be reserved until further evidence is considered.

[11]Variables 8 and 21 represent plateau-trial performances; they were therefore not included in the group discussed in the preceding paragraph.

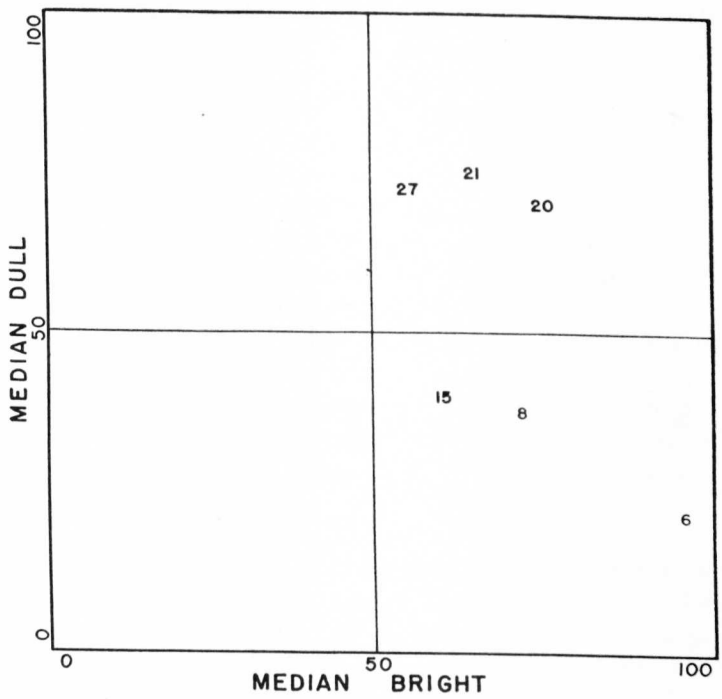


FIGURE 5*a*

ERROR MEASURES: FOOD PROBLEMS

D. MOTIVATION TO RUN FOR FOOD

The measures of starting time and total time of running relate more directly to motivational levels of the groups. These variables are plotted separately in Figure 5*b*. It may be noted by referring to the figure that whereas the points are scattered on both sides of the center vertical line, they are all in the lower half of the diagram, indicating that Dulls were universally low in these measures. It was observed on numerous occasions during the experiment that Dulls were typically apathetic towards food. They were slower to begin eating at the food box than were Brights and *MxM*'s and often failed to eat all of their daily ration.[12] On the whole, the evidence indicates that Dulls as a group are characteristically below average in food drive.

As regards Brights and *MxM*'s, no consistent differences were observed

[12]In this connection it may be recalled that the Dulls' weights are also lower.

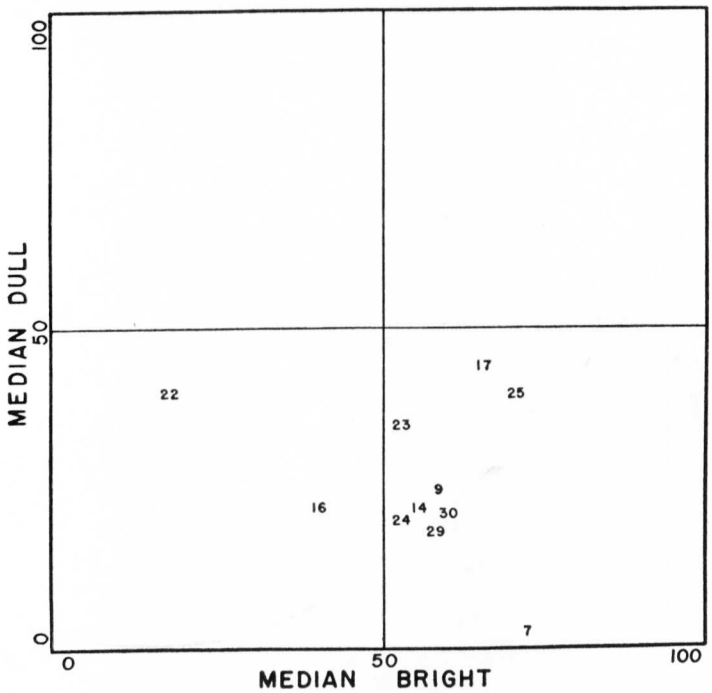


FIGURE 5*b*
TIME MEASURES: FOOD PROBLEMS

either in running speed or in eating behavior. The points in Figure 5*b* indicate that Brights' time scores were somewhat lower on the average (i.e., "better") than those of *MxM*'s, but these scores were influenced by various factors other than speed of running. For example, the marked influence of errors may be seen by comparing the positions of Variables 6 and 7, the measures of errors and time, respectively, on the Tryon Maze. Variable 7 would have appeared much closer to the mean *MxM* if corrected for number of errors. Insofar as strengths of food drive can be inferred from these sources of evidence, then, it may be supposed that Brights are no more strongly motivated by the food incentive than are average rats.[13]

[13]That is, in an absolute sense. As discussed in a later section, there are indications that food is stronger than other motivating factors for Brights and therefore is probably a more significant factor in their behavior as a whole than for other rats.

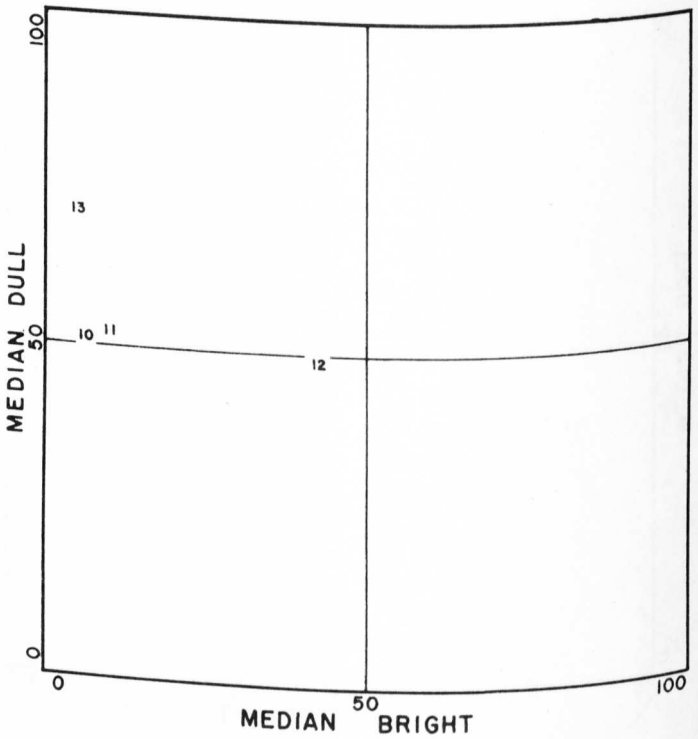


FIGURE 5*c*
TIME AND ERROR MEASURES: WATER PROBLEMS

E. ESCAPE-FROM-WATER MOTIVATION

In Figure 5*c* are shown the time and error measures which were taken for the Water Tank and the Water Unit. In contrast with their performances in food-reward problems, the Brights proved to be consistently poor in both time and errors, while Dulls were average or better. Although further study is needed in order to evaluate these results fully in relation to the rats' maze behavior, they indicate clearly that motivational differences of a complex character exist between the two strains. The fact that Brights' scores are correspondingly low in revolving cage activity (see discussion below) suggests that their performances here may be related to their low level of spontaneous energy.

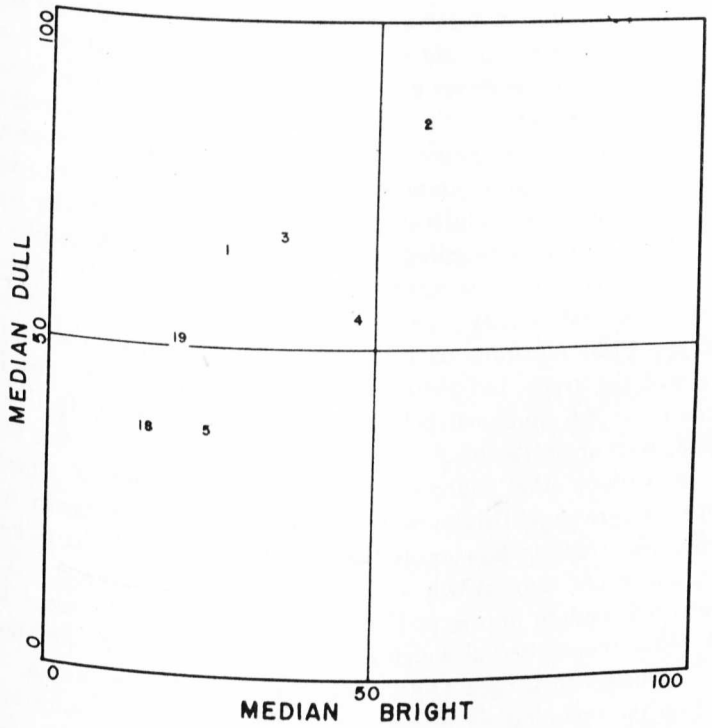


FIGURE 5*d*
OPEN FIELD AND ACTIVITY MEASURES

F. ACTIVITY AND EMOTIONALITY

A fourth selection of measures is shown in Figure 5*d*. These are Variables 1 to 5, inclusive, which were taken in the Open Field apparatus, and Variables 18 and 19 representing Revolving Cage activity.

In Revolving Cage activity, both pure strains were lower than the *MxM* group. Brights, however, were definitely less active than Dulls and, as may be seen by referring to the histogram of Variable 18 in Figure 4, even the most active of the individual Brights was well below the average *MxM*. The fact that the "exploratory distance" measure taken in the Open Field (Variable 5) showed such similar results suggests that both of these measures reflect the rats' general activity levels.

The Brights' position in Variable 19 is particularly significant, since this variable provides a direct indication of the relative strengths of their food and spontaneous activity drives in comparison with Dulls and *MxM*'s. As

there was no good criterion for evaluating the percentage scores in terms of high and low, the variable might have been placed on the opposite side of the graph with equal justification. However, its position indicates essentially that what little activity the Brights showed was due to hunger. Inspection of the raw scores reveals that the Brights were 60 per cent more active during the two hours before feeding than for the entire remainder of the daily period, while for Dulls and *MxM*'s the bulk of activity occurred at night. Thus, since pre-feeding scores are included in Variable 18, an even greater discrepancy exists in the spontaneous activity levels of the three strains than this total activity measure indicates.

The Open Field measures tend to scatter rather widely over the upper left section of the graph, but their position as a group signifies that Brights showed more of the emotional behaviors than average while Dulls, on the other hand, were more "stable." On the whole, this result agrees with evidence from various other sources. In subsequent running on some of the other types of apparatus, Brights were found to be characteristically stubborn during the initial trials, showing at the same time the teeth-chattering, face-washing, and other signs which are typical of emotional upset. This behavior was first noticed during preliminary training on the Elevated Path, in which Brights' time scores, although equal to those of *MxM*'s in later trials, were much higher than both Dulls' and *MxM*'s in the beginning. It also occurred on the two elevated mazes and on the 6-Unit Discrimination apparatus, each time being confined, however, to the early trials. It might be characterized, then, as a response to unfamiliar "open space" situations. Direct measures of the behavior were not obtained, but its influence on running times in each of the apparatuses was typically like that shown by the total time measures for the 16-Unit Maze. Referring to Figure 5*b*, it may be seen that Variable 22, times for the first two trials, deviates markedly to the left, while the positions of Variables 23 and 24, the plateau and final trial scores, respectively, are "normal." With regard to emotional responses of the Dulls, the only indications found in the present series of problems were incidental in nature. It was noticed during preliminary training on the 6-Unit Discrimination apparatus that the Dulls were very reluctant to step on a platform which was held by the experimenter (sometimes unsteadily) to aid their first crossings. A specific fear of unstable platforms seemed to be characteristic. In preparing to run final darkness trials of the 16-Unit maze, a plate switch was installed on the pathway at the "End" position (cf. Figure 3) for the rats to step on in passing. The plate was fixed to depress slightly with the rat's weight, turning on room

lights and thereby enabling total times to be recorded by stopwatch. Most of the rats stopped upon meeting the switch for the first time (at the end of Trial 14). The Dulls, however, were exceptionally upset by it and refused to cross until blocked from retracing by the experimenter. The effect carried over to the following trials (15 and 16), causing them to begin slowing up successively farther back into the maze on each trial. In Figure 6, which shows time curves for all three groups during the later trials, it is apparent that Dulls' total times were affected significantly.[14]

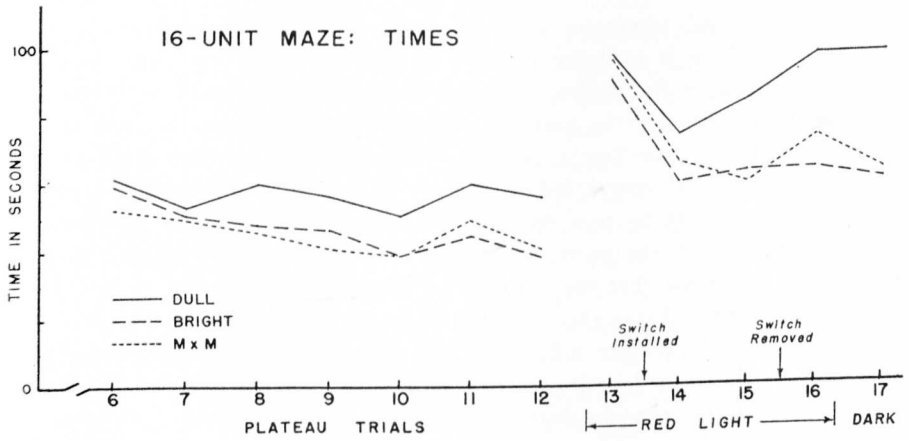


FIGURE 6
CURVES OF TOTAL TIMES FOR BRIGHTS, DULLS, AND *MxM*'S DURING PLATEAU AND FINAL DARKNESS TRIALS ON THE 16-UNIT ELEVATED MAZE

G. SUMMARY AND DISCUSSION OF THE ANALYSIS

The general findings of the preceding analysis may be summarized briefly as having indicated the following points:

1. Brights' scores on the 30 test measures tend to be slightly higher on the average than those of Dulls. However, this result depends upon selection of the particular tests which were included in the sample of test measures. With reference to the measures of errors in *learning,* where all types of the apparatus situations are considered, there is no evidence that Dulls are generally inferior.

2. As indicated by their running speeds and their behavior towards food, Dulls are below average in strength of food drive. No evidence was ob-

[14]The switch was removed just after Trial 15 but its absence was not discovered by the rats until the *end* of Trial 16.

tained which would permit differentiating between Brights and average rats with regard to food drive alone, but food appears to be more important for Brights in relation to other sources of motivation.

3. In problems involving escape-from-water motivation the Brights were consistently inferior, while Dulls were average or slightly better.

4. Both pure strains were below average in the Revolving Cage and Open Field measures of spontaneous activity. The Brights were particularly low, and showed the greatest *proportion* of their activity in the periods before feeding.

5. Different kinds of emotional responses appear to be characteristic of the two strains. There is evidence that Dulls have a specific fear of certain apparatus features, while Brights are initially timid in "open space" situations.

It remains to consider the possible relation of these various behaviors to the rats' hereditary maze-brightness and maze-dullness.

The finding in (1) above indicates that a "general intelligence" factor, if it exists at all, may be regarded as of little or no importance.[15] It was found, however, that the particular traits of each strain do appear to be "general" in the sense that they characterize the rats' behaviors in varying apparatus situations. From this, together with the intercorrelational evidence that Brights and Dulls are differently organized, it may be assumed that differences in the maze-learning ability represent differences in *patterns* of behavior traits rather than in *degrees* of any single psychological capacity.

1. *Brights*

The Brights' performances on all the learning problems where food was employed as an incentive may be taken to indicate that food drive is an important factor in their hereditary maze-brightness. It probably accounts to an appreciable extent for their superiority over Dulls. However, since the more direct indicators of motivation—running speed and behavior towards the food itself—showed no difference between the Brights and *MxM*'s, its importance seems to depend largely upon the relative strengths of other motivating factors.

Probably the most significant of the other factors is their exceptionally low spontaneous energy level. Both the Open Field distance measure and the 24-hour Revolving Cage score showed them to be lower than Dulls and *MxM*'s in general activity, while the percentage scores for pre-feeding sug-

[15] A similar result has been obtained consistently in previous correlational studies (7, 15, 22) and has been interpreted to mean that "specific factors" are largely responsible for individual differences in each type of apparatus.

gests that even this small amount should be attributed to hunger rather than to spontaneous energy. One might predict on logical grounds that rats having a motivational structure of this sort would show a "least distance" tendency in the alley maze, food-reward situation: being stimulated almost exclusively by the need for food and having little excess energy, they should be less distractible, should eliminate blind alleys more readily, and on the whole, should show behavior of a more "purposive" character than other rats.[16] In general, these expectations are consistent with Tryon's findings (19, p. 319) that Brights are relatively more controlled by the food-pointing, exit gradient, and higher direction-configuration components of the maze.

It may, however, be too easy to overestimate the importance of the food-activity balance in the Brights' total organization at the expense of other traits which have not been adequately identified. The "initial timidity" characteristic, for example, is potentially significant but its relation to maze error scores is not yet clear. The problem is somewhat complicated by apparent differences which have been observed as between the elevated and the alley maze situations. Tryon, Tryon, and Kuznets (20, 21) found that Brights were *less* emotional in their reactions to doors, curtains, alleys, and floors in the alley maze, but that in reaction to handling by the experimenter they showed more of the "avoidance" and "escape" responses. The latter finding suggests a similarity to the "open space" timidity shown in the present experiment, although handling was not always involved in this case. On the whole, the evidence indicates that the Brights' timidity is related to handling, to open-space and elevated types of apparatus, and to strangeness of the situation; but further studies are obviously needed before any precise definition of this trait should be attempted.

2. *Dulls*

In the measures of learning in food problems, Dulls showed no consistent tendency to be either high or low in errors, but their speeds of running were definitely below average. Considering the other indications that Dulls are generally low in food drive, the assumption is probably justified that low motivation accounts in part for their performances in the alley maze. However, since their errors in the other food problems were equal to those of Brights the importance of this factor should not be overemphasized. Either

[16]This hypothesis is not novel. Washburn, in an early experiment with white mice, compared error scores of the predominantly activity-driven mice with those of the predominantly hunger-driven and found that fewer errors were made by the latter group (23).

its influence upon errors is very slight, or other factors were operating in the elevated apparatus to compensate for its effect.

The indication that Dulls have a specific fear of unstable platforms should be studied further. It is conceivable that non-retrace gates and movable floors in the Tryon Maze would normally produce effects similar to those of the automatic light switch which was used during final trials of the 16-Unit Maze. Since non-retrace gates are always on the true path side of each choice point (necessarily), the effect of avoiding them would be an increase in errors as well as in total times. The implication may be that Dulls actually do learn the true path but that their learning, paradoxically, is manifested as entrances into, rather than as avoidance of, blind alleys. The various other facts which are known regarding the Dulls' psychological organization—namely, (*a*) their demonstrated ability to learn complex problems during the present experiment, (*b*) the indication of their unusually low food motivation, and (c) Tryon's finding that they show emotional reactions in the maze itself—are uniformly consistent with such a hypothesis.[17]

[17]Direct evidence in the Tryon Maze might be obtained by making systematic changes of movable floors and non-retrace gates, replacing them with devices of a different design. Dulls still might not perform as well as the Brights, but some improvement should be noticed if the hypothesis is correct.

V. GENERAL SUMMARY

An exploratory study of behavior organization in the Tryon strains of maze-bright and maze-dull rats was conducted with the purpose of determining the nature of psychological factors in their hereditary maze intelligence. As a result of preliminary studies in which the writer found that the two strains did not differ significantly in their abilities to learn an abstract spatial pattern, the hypothesis was suggested that Brights and Dulls are different psychological "types" of rats. The present experiment was designed to test this hypothesis and to identify the specific behavior tendencies which are thought to enter into each strain's maze-learning performance.

Thirty measures were obtained of learning, emotionality, activity, and other behaviors for samples of 10 Brights and 10 Dulls, and for a sample of 15 rats of a median strain. The latter group was taken as a reference group representing the normal rat population. The raw scores of rats in the two experimental groups were converted into standard scale values based upon scores of the reference group in each variable and these were used to obtain behavior profiles for members of the Bright and Dull strains.

As a modified form of the "inverted" factor analysis method, correlations were computed between the obtained score distributions for rats in order to determine the degree of homogeneity within each strain and the extent of similarity between strains. The mean correlation of Brights with Dulls was found to be low and negative (—.19), whereas members within each strain showed relatively high and positive correlations with one another (+.59, +.53). Although conclusions are conditioned to some degree by unreliability caused by the use of a small reference group, these results strongly confirm the type hypothesis.

No evidence was found that a difference exists between the Brights and Dulls in the learning capacity *per se*. A detailed study of the behavior profiles indicated that Brights are characteristically food-driven, economical of distance, low in motivation to escape from water, and timid in response to open spaces. Dulls are relatively disinterested in food, average or better in water motivation, and timid of mechanical apparatus features. It is concluded that brightness and dullness in the original Tryon Maze may be accounted for in large part by such motivational and emotional patterns. Although indications exist that the two strains may also be differentiated with reference to certain basic "cognitive" tendencies, the procedures followed in this experiment were not sufficiently analytical to indicate their nature. The need for further study is evident, both to relate the factors identified to the maze-learning performance in particular, and to discover other factors which may be present.

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